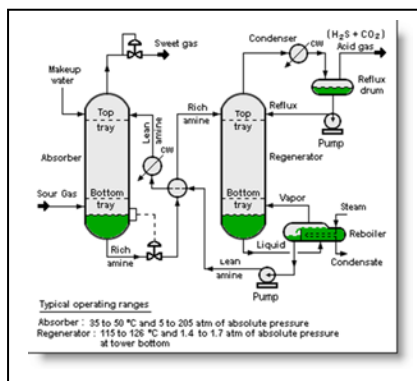


UNIVERSITY OF ILLINOIS AT CHICAGO

Department of Chemical Engineering



College:
Department:
Course:
Time and Place:

UIC COLLEGE OF ENGINEERING
UIC Department of Chemical Engineering
CHE 433 Process Simulation with ASPEN Plus
SES 138 MW: 09:30 A.M. – 10:45 A.M. F: 10:00 A.M. – 10:50 A.M.

Instructor:
Office:
E-mail:
Cell Phone:

Dr. Michael Caracotsios
EIB 244
mcacarot@uic.edu
(630) 696-0934

COURSE OBJECTIVES: This course presents the main principles of Process Simulation (Material and Energy Balances) relevant to chemical and biological processes. Process Simulation or Flowsheeting is usually defined as the use of commercially available products (like ASPEN Plus, ASPEN Hysys, UniSim, VMG, ChemCad, gProms, Pro II, etc.) to perform steady-state Heat and Weight Balance as well Sizing and Cost calculations for chemical, biological, petrochemical and refining processes. By the end of the semester the students should be able to:

- ✚ Select and justify choices for physical property models required for accurate heat and weight balance calculations.
- ✚ Design and model unit operations such as chemical reactors, heat exchangers, heaters and coolers, pumps, compressors and turbines, mixers, distillation and packed columns and user specific operations.
- ✚ Combine units operations in a cohesive Flowsheet that describes existing or conceptual processes.
- ✚ Familiarize themselves with computational aspects such as design specs, calculator blocks, sensitivity analyses, flow sheet convergence
- ✚ Perform optimization studies.

COURSE PREREQUISITES: The following are key requirements for a successful completion of this course.

- **Integral and Differential Calculus.**
- **Multivariable Differential Calculus**
- **Familiarity with a Programming Language.**
- **CHE Department Prerequisites: CHE 301, CHE 311, CHE 312, CHE 313 and CHE 321.**

SOFTWARE: Students may download [FREE of charge] the software package **Athena Visual Studio** from www.AthenaVisual.com. Students will also use EXCEL, ASPEN Plus and optionally MATLAB and ASPEN Hysys.

RECOMMENDED TEXTBOOKS: [Textbooks are Optional. My Personal Notes will be distributed to All Students]:

1. Ralph Schefflan, *Teach yourself the Basics of Aspen Plus*, 2nd Edition, Wiley. (2016).
2. Gavin Towler and Ray Sinnott, *Chemical Engineering Design*, 2nd Edition, Elsevier Ltd. (2013).
3. Lorenz T. Biegler, Ignacio E. Grossmann and Arthur W. Westerberg, *Systematic Methods of Chemical Process Design*, Prentice Hall International Series, December (1999).
4. T. F. Edgar and D. M. Himmelblau, *Optimization of Chemical Processes*, McGraw-Hill, Inc.(1988).
5. A. W. Westerberg, H. P. Hutchison, R. L. Motard and P. Winter, *Process Flowsheeting*, 1st Edition, Cambridge University Press (1979).

COURSE OUTLINE: CHE 433 PROCESS SIMULATION WITH ASPEN PLUS.

1. REVIEW AND BACKGROUND

- Introduction to Process Flowsheeting in Chemical Engineering
- Elements of Differential and Integral Calculus
- Numerical Methods for Algebraic and Differential Equations
- Mass and Energy Balances in the Context of Process Simulation
- Tutorial on Programming with ASPEN Plus and Athena Visual Studio

2. THERMODYNAMICS AND PROCESS UNIT OPERATIONS

- Vapor-Liquid Equilibrium and Property Methods
- VLE Data Regression
- Mixers and Splitters
- Two and Three Phase Flash Blocks
- Heaters, Coolers and Condensers
- Shortcut and Rigorous Distillation
- Pumps and Compressors and Turbines
- Example Flowsheets and Design Specifications

3. PROCESS FLOWSHEETING FUNDAMENTALS

- Recycle Stream Selection and Tear Equations
- Convergence Methods for Tear Equations
- Calculator and Transfer Function Blocks
- Simultaneous Recycle Convergence and Design Specifications
- Sensitivity Analysis and Case Studies
- Sequential Modular vs. Equation Based Solution Methods

4. CHEMICAL KINETICS AND REACTOR UNIT OPERATIONS

- CSTR, Plug Flow and Fixed Bed Reactors
- Conversion and Equilibrium Reactors
- Example Flowsheets with Design Specifications
- Excel Interface for Custom Reactor and Physical Models

5. COMPLEX UNIT OPERATIONS AND ADVANCED SIMULATION TOPICS

- Packed columns for Absorption and Stripping Operations
- Rate based vs. Equilibrium Models
- Homogeneous and Heterogeneous Azeotropic Distillation
- Process Optimization and Pinch Analysis for Energy Optimization
- Heat Exchangers and Heat Exchanger Network Optimization
- Solids Handling and Electrolyte Systems

ACADEMIC INTEGRITY:

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy:

<https://dos.uic.edu/conductforstudents.shtml>

RELIGIOUS HOLIDAYS:

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure.

The University Holidays and the Religious Observances calendar can be found online at the following website: <http://oae.uic.edu/religious-calendar/>. The list includes national holidays recognized by the University as well as religious days of special observance that may prohibit a student from attending classes. Please keep in mind this list is not exhaustive.

PARTIAL CREDIT POLICY:

Partial credit can be requested under certain circumstances. The student should make an appointment with the Professor and be prepared to discuss the reasons for requesting the extra credit. Final decision rests with the Professor.

Partial credit can also be requested in cases where there is an obvious (indisputable) error made by the Professor or the Teaching Assistant.

No partial credit will be given for:

1. Incorrect formulas, full or partial.
2. Inconsistent or missing units in calculations.
3. Multiple answers.
4. Incoherent (illogical, disjointed) derivations or calculations.

STUDENT COURTESY POLICY

Talking/Smart Phones/Laptops/Food and Drinks in Class:

Not allowed unless explicitly stated

Repeat Offenders will be asked to leave the classroom

Cheating/Copying: The entire class gets penalized after the first warning

Homework contribution to the final grade is substantially reduced

Perpetrators will receive zero points in their homework

Late Homework and Disputes:

Late Homework will not be accepted without prior arrangements with your Professor in person

If there are Homework and Exam score disputes they will be resolved only with the Professor.